

PHASE 2A · COST AND CONTRACT INTELLIGENCE

EngiSignal Phase 2A

Phase 1 closed with seven of eight capabilities reachable and the eighth — the one that turns an observation into a purchasing decision — structurally unreachable. This phase closes the financial loop, and finding two defects in the process that only money could have exposed.

Deployment engi-signal.vercel.app Commit 88c3cd7 Verified 15 Aug 2026
Scope Contracts & cost only

ESLINT	TYPECHECK	TESTS	PROD BUILD
0 problems	0 errors	507 / 507	Pass
RLS COVERAGE			
29 / 29			

A What was implemented

A fourth first-class dataset through the *existing* pipeline. No second parser, mapping engine, validation framework or analytics engine was created — `lib/analytics/financial.ts` was not modified at all, because it was already correct and merely starving.

SHIPPED

Contracts & cost import

Upload → detect → map → validate → preview → import → verify, identical to the other three datasets. CSV, TSV, XLSX and XLSM, single- and multi-sheet. Legacy `.xls` remains refused, as the

parser genuinely cannot read it.

SHIPPED

Deterministic cost derivation

`lib/ingestion/cost.ts`. Every figure carries the basis it was derived from, stored beside the fields that produced it so it can be recomputed by hand.

SHIPPED

Conservative contract-to-feature matching

`lib/ingestion/contract-match.ts`. Exact normalized name, vouched-for SKU, or confirmed alias. Nothing else links, and everything that does not becomes a review item naming both sides.

SHIPPED

Renewal exposure

`lib/analytics/renewal.ts`. Cumulative 30/60/90/180/365-day windows, with undated and lapsed lines reported separately rather than bucketed by assumption.

SHIPPED

Renewal positions table

One row per feature on the existing Renewals page: owned, annual cost, recommended, cost at recommended, opportunity, renewal timing, and an evidence phrase stating what the figures rest on.

UNCHANGED

Signals

No new signal code. The deterministic engine already produced cost, renewal and reclaim signals from `PortfolioRow.financial`; supplying real prices was all it needed. Verified live: *"\$485K optimization opportunity identified before commitment"* and *"Ansys Mechanical is over-provisioned — entitled capacity exceeds demand by 97 licenses."*

B

Database changes

One new table. `contracts` and `contract_items` were deliberately *not* written into: they are analytics-grain, keyed on `feature_id` and `vendor_id`, with `renewal_date NOT NULL`. A renewal spreadsheet satisfies none of that on arrival, and forcing it to would mean guessing an identity at `INSERT` time and inventing a date for perpetual

licences.

CHANGE	DETAIL
New table	<code>ingestion_contracts</code> — 28 columns at source grain, every raw string preserved, full provenance, RLS enabled and forced , four policies matching the other ingestion tables exactly.
Enum	<code>canonical_dataset</code> gains <code>contracts</code> .
Column	<code>imports.contract_records</code> , so the accepted/rejected reconciliation stays complete for this dataset.
Constraints	Non-negative money; currency must match <code>^[A-Z]{3}\$</code> ; contract end cannot precede start; basis columns constrained to the six known values.
Indexes	Unique on <code>(import_id, source_sheet, source_row)</code> ; lookup on <code>(organization_id, raw_feature)</code> ; partial on <code>(organization_id, renewal_date)</code> for exposure scans.

Deleting an import remains `delete ... where import_id = $1`. Verified in production: removing the contracts import cleared 13 commercial rows and 6 rejection records while leaving all 40 usage rows untouched.

c Canonical schema

`CanonicalContractRecord` — identity, commercial terms, dates, context, derivation, provenance.

GROUP	FIELDS
Identity	<code>feature, product, vendor, sku, contractNumber, agreementNumber, purchaseOrder, supplier</code>
Commercial	<code>quantity, unitPrice, totalCost, annualCost, currency, licenseModel, pricingUnit</code>
Dates	<code>contractStartDate, contractEndDate, renewalDate</code>
Context	<code>businessUnit, costCenter, owner, notes</code>

Derivation	unitPriceBasis, annualCostBasis, multiYearTotal
Provenance	organizationId, importId, importedAt, sourceFile, sourceSheet, sourceRow, sourceSystem

THE MINIMUM IMPORTABLE RECORD, STATED EXPLICITLY

A row must carry **an identity** — `feature`, or a `product` or `sku` standing in for it — **and at least one of** a cost basis (`unitPrice`, `totalCost`, `annualCost`) **or** a contract date (`renewalDate`, `contractEndDate`).

The second half is deliberately not "must have a price". A renewal schedule that lists dates and leaves pricing to procurement is a real and useful document — it unlocks renewal exposure with no money in it at all. What the rule rejects is a row carrying a name and nothing else, which cannot reach any analysis and would otherwise pad the accepted count with rows that do nothing.

D Mapping logic

The Phase 1 resolver, unchanged, with a commercial alias table layered on. Confidence and a real sample value are shown for every proposal, and every one is editable before commit. A customer imports a messy renewal spreadsheet without renaming anything.

THE MONEY COLUMNS NEEDED CARE

`Price`, `Cost`, `Total` and `Spend` are all ambiguous alone, and reading a line total as a unit price overstates a position *by the quantity* — a hundredfold error on a 100-seat line. Each bare word is claimed exactly once, by the field it most often means on a purchasing document: `price` → unit price, `cost` → total cost, `spend` → annual cost. Qualified spellings such as `unit_cost` or `annual_price` match their own field exactly and win outright.

Verified against a synthetic procurement file with none of EngiSignal's vocabulary:

`Publisher` → vendor, `Software Name` → product, `SKU Description` → feature, `Part`

Number →SKU, Qty →quantity, Unit Price →unitPrice, Annual L
Spend →annualCost, Total →totalCost, Renewal Date →renewalDate,
PO →purchaseOrder, License Type →licenseModel. A second file using only
Expiration Date mapped it to renewal date, which is the column a renewal
schedule exists to communicate.

E

Financial formulas

QUANTITY	RULE
Unit price	Supplied value wins. Else annual ÷ quantity. Else total ÷ quantity, only when the term is one year or less. Else unknown.
Annual cost	Supplied value wins, <i>even when it disagrees with quantity × unit price</i> — the disagreement is the customer's to explain and must not be hidden by preferring our own arithmetic. Else quantity × unit price. Else the total, when the term is a year or less.
Merged position	Quantities add across purchase orders. Unit price is quantity-weighted : \$4,000 across 10 seats and \$6,000 across 990 averages to \$5,980, not \$5,000 — a naive mean would misprice that position by about \$980,000.
Recommended	<code>ceil(P95 × growth × safety)</code> — Phase 1's engine, unmodified.
Surplus	<code>max(entitled - recommended, 0)</code>
Opportunity	<code>surplus × unit price</code> , and only where demand was actually observed.
Renewal exposure	Cumulative windows. A line at 45 days appears in 60, 90, 180 and 365, because it is exposure inside all of them.

THE RULE DELIBERATELY ABSENT

A multi-year total is never divided by its term. A \$2.7M line against a three-year contract "obviously" annualizes to \$900,000 — but a Total column may be a one-year price, a three-year prepayment, a co-termed true-up, or a ramped deal whose years are not equal. Straight-lining it produces a confident annual figure the customer never agreed to, and a position built on it collapses the

moment the vendor produces the actual schedule. Such lines are flagged, carried as `totalCost`, and left out of annual reporting with the reason stated.

The worked example, verified in production

275	P95 of daily peak demand, from 20 days of imported FlexNet usage
× 1.00	Growth factor — no headcount growth rate was supplied, so none is assumed
× 1.10	Safety buffer, the configured default
= 303	302.5 rounded up to a whole licence
400	Entitled quantity, from the contract line
97	Surplus: 400 – 303
\$5,000	Unit price, stated in the imported file — basis <code>supplied_unit_price</code>
\$485,000	Annual opportunity: 97 × \$5,000

Every number above traces to a row in a file the customer supplied. Nothing is benchmarked, inferred or defaulted.

F Matching rules

A contract line links to a feature on one of exactly three grounds:

- 1. **Identical normalized name.** The same rule usage and entitlements already use — case and separators only.
- 2. **SKU equality.** but only where another line already tied that SKU to a feature by

also matching on name. It propagates a justified link; it never manufactures one.

3. **A customer-confirmed alias.** A human answered the question.

WHY NOTHING FUZZY

A similarity score would happily merge `ansys_mech_ent` at \$4,000 with `ansys_mech_premium` at \$9,500. They share a long prefix and differ by one word. Merged, the blended price is wrong for both, the entitlement is wrong for both, and the recommendation — surrender 80 licences — is presented with full confidence. The customer takes it into a negotiation and is shown by their vendor that it is nonsense. One such failure costs more trust than fuzzy matching could earn back in convenience.

An unmatched line is **not discarded**. It still counts toward spend and renewal exposure — dropping it would understate the portfolio by exactly the amount nobody has reconciled yet, which is the amount most worth seeing. What it does not get is a comparison against demand, because there is none. It appears in the review queue with the lookalikes named as suggestions and a sentence saying what would settle it.

G Capabilities unlocked

CAPABILITY	REQUIRES	STATUS
Financial opportunity	usage + entitlements + cost	NOW REACHABLE
Recommended renewal spend	usage + concurrency + entitlements + cost, ≥14 days	NEW
Renewal exposure	contract dates alone	NEW · VERIFIED
Reclaim opportunity	usage + people + named-user + cost	NEW · NOT YET EXERCISED
The same Phase 1	unchanged	unchanged

The seven Phase 1 capabilities	unchanged	UNCHANGED
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Renewal exposure deliberately needs no usage at all: a customer who uploads only a renewal schedule gets a working calendar on their first import rather than a locked screen. Recommended renewal spend inherits the 14-day history floor — a percentile from four days does not become defensible by having a dollar sign in front of it.

H

What is still unavailable

DATA MISSING	Cost per active user, cost per utilized licence Both functions exist and are correct. They need resolved people, which needs a People import. Not exercised this phase.
DATA MISSING	Reclaim value for named-user licences Requires people, named-user licensing and cost together. The capability gates correctly and was confirmed withheld for a purely concurrent estate, where "inactive seat" has no meaning.
DATA MISSING	Cost-centre and business-unit allocation Both fields are imported and stored. No allocation view was built on them, because allocation without resolved people would attribute spend on incomplete evidence.
BY DESIGN	Cross-currency totals A position spanning two currencies is left unpriced and reported as mixed. Summing without a rate would be arithmetic on incomparable units, and EngiSignal holds no FX rates.

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Security results

VERTETER	Tenant isolation on the new table
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VERIFIED	tenant isolation on the new table RLS enabled and forced on <code>ingestion_contracts</code> with all four policies, identical in shape to the existing ingestion tables. All 29 public tables now carry RLS. Supabase advisors report no finding against it.
VERIFIED	Organization never taken from the request Resolved server-side from authenticated membership, unchanged from Phase 1. Records whose provenance names a different organization are refused at commit rather than rewritten — covered by test.
VERIFIED	Cross-tenant read and delete A foreign import id returns <code>404</code> in production. Automated tests cover cross-tenant read, cross-tenant delete and mismatched provenance for the commercial dataset specifically.
VERIFIED	No service-role key Still none. Every database call runs as the signed-in user.
REVIEWED	<code>bootstrap_organization</code> advisory Unchanged from Phase 1D and still accepted: the escalation is bounded, the owner derives from <code>auth.uid()</code> , the role is fixed rather than accepted, and <code>search_path</code> is pinned.
OPEN	Leaked-password protection is disabled Supabase can check passwords against HaveIBeenPwned and currently does not. This is a project setting rather than code, so it was not changed here — worth enabling before external users, and it costs nothing.

QA residue

One temporary account was created for production verification and deleted afterwards. Users, organizations, memberships, imports, canonical usage, **canonical contracts**, entitlements, people, rejections, features, contracts and contract items all return **zero rows**. No QA credentials, emails, organization ids or import ids exist anywhere in the codebase.

J Automated test results

507 passing, up from 432. All 432 Phase 1 tests still pass unmodified.

AREA	TESTS	COVERS
Commercial mapping	4	Real-world headers, expiry-only schedules, bare-money-word collisions, reviewer override
Validation	8	Row reconciliation, no-commercial-content, invalid currency, negative price, inconsistent dates, unreadable date, duplicates, and the two "accepted but incomplete" cases
Cost derivation	9	quantity × price, explicit annual preferred over arithmetic, annual ÷ quantity, one-year total, multi-year refusal, short-term no-scaling, missing quantity, all-missing, derivation recording
Currency	4	ISO codes, unambiguous symbols, bare "\$" refused, absent vs unreadable
Matching	5	Name, alias, SKU propagation, two-SKU non-merge, unvouched SKU refused
Position merge	4	Quantities add, quantity-weighted price, earliest renewal binds, mixed currency unpriced
End-to-end financial	5	Priced surplus, unpriced stays null, entitlement precedence, review surfacing, contracts-only portfolio
Renewal exposure	4	Cumulative windows, unpriced reported, undated excluded, unknown ≠ low urgency
Capability gating	7	Financial withheld then unlocked, exposure from dates alone, reclaim needs named-user, missing ≠ zero, mixed currency warning
No-evidence guard	3	The defect in section L — no surrender without demand, evidenced features still sized, totals exclude unevidenced lines
Detection	2	Commercial files not attributed to a vendor tool; explicit source still honoured

Persistence & security	14	Storage, traceability, derivation storage, per-import deletion, coverage, duplicates, fingerprinting, cross-tenant read/delete/provenance
Workbooks	3	Multi-sheet XLSX, XLSM, TSV
Phase 2A regression	3	Usage analytics byte-identical with and without contracts, including when a line matches nothing

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Production tests performed

TEST	RESULT	EVIDENCE
Authenticated usage import	PASS	40 rows, 20 days, P90/P95/P99 unlocked
Messy CSV contract import	PASS	13 accepted + 6 rejected = 19 read; all six rejection rules fired with example values
Multi-sheet XLSX contract import	PASS	Sheets FY26 + FY27, 3 lines, prices and dates correct
Financial capability unlock	PASS	\$485K opportunity with the full derivation in section E
Renewal exposure	PASS	180d: 1 line \$2.0M; 365d: 3 lines \$2.7M; undated line excluded and reported
Deterministic signals	PASS	Cost and renewal signals with entitled/P95/recommended/opportunity shown
Supabase persistence	PASS	Rows in <code>ingestion_contracts</code> with basis columns populated
Sign out and back in	PASS	Both imports and all analytics intact
Duplicate contract upload	PASS	Refused; still 2 imports and 3 contract rows, no double-count
Per-import deletion	PASS	13 contract rows and 6 rejections removed; 40 usage rows untouched

rows unrendered		
Foreign import id	PASS	404
Console errors	PASS	Zero across app pages
Desktop 1440 / mobile 390	PASS	No horizontal page overflow; wide tables scroll in their own containers
Post-teardown residue	PASS	Zero rows across all 13 checked tables

L Bugs found and fixed

THE ONE THAT MATTERED – FOUND ONLY BY PUTTING PRICES IN

The production portfolio recommended reducing CATIA from 90 licences to 0 and banked it as **\$1.08M of "opportunity"** — because no usage had been imported for CATIA at all.

`computeConcurrentMetrics` returns a fully populated result when it has seen nothing: every figure zero, `observedDays` zero. The right-sizing gate tested `metrics !== null`, which is true in that state, so the percentile model ran on an empty array of daily peaks. Basis zero, recommended zero, surplus equal to the entire entitlement.

Phase 1 could not have caught this. No import path supplied a price, so unpriced lines produced no financial figure and the zero was invisible. Attaching real contract prices turned a latent arithmetic artefact into a confident, dollar-valued instruction to cancel a licence pool — absence of evidence sold as evidence of absence, at full contract value, on the one screen a customer takes into a negotiation. The gate is now `observedDays > 0`; cost is still reported, only the recommendation is withheld.

FIXED

Stated unit price silently discarded

Position merge reconstructed unit price only from annual cost, so a line stating \$5,000/seat but carrying no quantity went unpriced while the file plainly said otherwise. Now falls back to a quantity-

...into the the primary and other tabs. Now this back to a quality weighted average, then to a single stated price; two different prices with no quantities stay unpriced rather than being averaged unweighted.

FIXED

The review queue could never fire

Contract lines fed the same feature-discovery list they were then matched against, so every line matched itself by name, the unmatched path was unreachable, and the review queue was permanently and silently empty — defeating the entire conservative-matching design. Matching now runs against features observed in usage and entitlements only.

FIXED

Procurement files attributed to a vendor's tool

Detection scores vendor terminology, and a renewal schedule is a list of vendors by construction. A synthetic file was reported as "Dassault Systèmes DSLS · 60%" and every row stamped `sourceSystem: dsls` — a false claim in the one field whose job is to say where a number came from. Commercial files are now read as commercial exports regardless of which vendors they name.

FIXED

"Demand observed" claimed for features with none

Same root cause as the main defect, surfacing in the renewal evidence column: eight of eleven positions claimed observed demand that did not exist. Now reads "Priced, but no usage was imported for this feature."

M

Known limitations

- **Scale is unproven.** The largest production commercial import was 19 rows. Configured limits remain 500,000 rows and 25 MB, untested at anything near that.
- **No FX conversion.** Mixed-currency positions are reported, not totalled.
- **Alias confirmation has no dedicated UI.** Unmatched contract lines land in the existing unmapped-features queue; confirming one at scale has not been exercised.

- **Entitlement precedence is a judgement.** Where an entitlement and a contract disagree on quantity, the entitlement wins for utilization — the server's number is what demand was actually measured against. The disagreement is preserved but not yet surfaced as its own finding, and over-deployment versus shelfware is worth showing explicitly.
- **Utilization still reads 0% for unevidenced features.** Less dangerous than the recommendation defect and carrying a Low confidence badge, but the same class of statement; worth the same treatment.
- **Only FlexNet-shaped usage has met production.** Unchanged from Phase 1: RLM, DSLS and Sentinel remain fixture-tested only.
- **Transactional email is still unconfigured,** carried forward from Phase 1.

N

Deployment verified

Commit	88c3cd7a0d4a917cf5c26447c45028d763f77efb
Deployment	dpl_C12SxrnRQNEmT3pSeUw3XvsLuzDo
URL	engi-signal.vercel.app